

Skema Analisis Obligasi

Data Taxonomy, Analysis Pattern & AI Agent Blueprint

Reverse-engineered from 5 sample files (3 inputs, 2 target outputs),
with a worked agent run on PT Daaz Bara Lestari Tbk (LK interim 30 Juni 2026)

Isi / Contents

- 1 What the five files are — and how they chain
- 2 Data taxonomy (5 layers, field & ratio dictionary)
- 3 Data-quality & methodology issues the agent must handle
- 4 The analysis pattern (reasoning chain, rating logic, templates)
- 5 AI agent blueprint (architecture, contracts, prompt, QA)
- 6 Worked run: Analisis Rasio PT Daaz Bara Lestari Tbk
- A Lampiran: ekstraksi line item & jejak perhitungan

BAGIAN 1

What the five files are — and how they chain

The five files are not five independent documents. They are snapshots of one pipeline taken at different stages. Reading them side by side shows a clear flow: a raw financial statement is turned into a row of ratios, the ratios are judged against a fixed rulebook, event information and the investor's holdings are layered on top, and the result is written up as two kinds of report.

File	Layer	What it contains	Entity / date	Role for the agent
input_skema_analisis.pdf (86 hal.)	L1 · Raw source	Laporan keuangan konsolidasian interim (neraca, laba rugi, perubahan ekuitas, arus kas, 33 catatan). Teks dapat diekstrak; 2 halaman surat pernyataan direksi berupa gambar.	PT Daaz Bara Lestari Tbk · 30 Jun 2026 (tidak diaudit)	Source of absolute line items + qualitative facts (bond terms, rating, trustee, bank covenants, subsequent events).
input_2_skema_analisis.xlsx	L2 · Ratio row	1 sheet "Data", 1 row, 33 columns: identitas emiten + Z-Score, Modified Z, T1-T4 dan 26 rasio pendukung. Tanggal disimpan sebagai serial Excel 46203 (= 30 Jun 2026).	PT Pegadaian (PPGD) · 30 Jun 2026	Pre-computed ratio vector exported from the DIFAPro "Rasio Keuangan" screen.
input_3_skema_analisis.pdf (15 slide)	L3 · Rulebook	Materi DIFAPro: definisi & rumus rasio, model Altman Z-Score (EM), Modified Z-Score, daerah batas, rasio covenant per sektor, indikator pendukung.	PT Edvisor Profina Visindo · 2026	The scoring criteria. Every RED/YELLOW/GREEN in the outputs is traced back to this file.
output_skema_analisis_2.docx	L5 · Report A	"Analisis Menyeluruh Rasio Keuangan" — penilaian per area, peta risiko, trigger eskalasi, data yang harus diminta.	PT Pegadaian · posisi Jun 2026 + overlay Sep 2026	Target format #1: issuer credit assessment.
output_skema_analisis_.docx	L5 · Report B	"Analisis Exposure YKPP" — posisi kepemilikan sukuk/obligasi, implikasi, pertanyaan ke Direktur Keuangan & Wali Amanat.	YKPP holding Rp13 M in PPGD · Sep 2026	Target format #2: investor exposure memo + engagement questions. Builds on Report A.

Key finding. The raw statement (input_1) belongs to a **different company** (DAAZ) from the ratio row and the outputs (Pegadaian). So the samples show the full chain only from Layer 2 onwards. That gap is exactly what weakens the Pegadaian reports: Report A says four separate times that it *cannot* compute a pro-forma DER, Modified Z-Score or loss-absorption capacity because the ratio file has no absolute values (\$6, \$8, \$9, \$13). An agent that ingests Layer-1 statements like input_1 closes that gap. Section 6 of this document demonstrates it on DAAZ.



BAGIAN 2

Data taxonomy

The taxonomy has five layers. Each layer has a fixed vocabulary, so the agent can map any new issuer into the same slots. Below, every term is shown with its exact Indonesian label as it appears in the source files, because the agent must match on those labels when parsing.

2.1 Layer 1 — canonical line items (extracted from the LK)

These 22 fields are the minimum set needed to recompute all 33 ratios in input_2 and every threshold in input_3. Interim statements need a period flag so flow items can be annualised.

Field ID	Label in LK (contoh DAAZ)	Statement	Feeds
TA	JUMLAH ASET	Neraca	T1-T3, MZ, ROA, Debt Ratio
CA / CL	Jumlah Aset Lancar / Jumlah Liabilitas Jangka Pendek	Neraca	Working capital, CR, Quick
TL / TE	JUMLAH LIABILITAS / JUMLAH EKUITAS	Neraca	DER, T4, SOLVR, ROE
RE	Saldo laba (ditentukan + belum ditentukan penggunaannya)	Neraca	T2
INV	Persediaan	Neraca	Quick Ratio
AR	Piutang usaha (pihak berelasi + pihak ketiga)	Neraca	Piutang/Aset Lancar
CASH	Kas dan bank	Neraca	Net Gearing
IBD	Utang bank (pendek + jatuh tempo 1 thn + panjang) + Utang obligasi	Neraca + Cat. 11, 13	Net Gearing, IBD/E covenant, DSCR
LTD	Utang bank jangka panjang + Utang obligasi jangka panjang	Neraca	LTD/E, LTD/TA
REV / COGS / GP	Pendapatan / Beban pokok pendapatan / Laba kotor	Laba rugi	Margins, T5, growth
OPEX → OI	Beban usaha → Laba operasional = Laba kotor – Beban usaha	Laba rugi	ICR, OPM, OPPROF_TA
INT	Beban keuangan	Laba rugi	ICR, EBITDA/Interest, DSCR
NI	Laba bersih periode berjalan	Laba rugi	NPM, ROA, ROE, CASHPROF
D&A;	Beban penyusutan (+ amortisasi)	Cat. aset tetap	EBITDA, CASHPROF
CFO	Kas bersih dari aktivitas operasi	Arus kas	OCF/NI, Cash Flow Coverage
PRIOR_*	Pembanding periode sama tahun lalu	Laba rugi / Arus kas	Growth (YoY)
PERIOD	Tahunan / Semester / Kuartal + tanggal	Header	Annualisation factor
QUAL_BOND	Seri, nominal, kupon, tenor, jatuh tempo, rating, wali amanat	Cat. utang obligasi	Report B, maturity wall
QUAL_COV	Covenant pinjaman bank / perwaliamanatan	Cat. utang bank	Covenant headroom
QUAL_EVENT	Peristiwa setelah periode pelaporan	Cat. terakhir	Overlay / positive evidence
QUAL_AUDIT	Opini auditor / status (diaudit/tidak)	Laporan auditor	Indikator pendukung (WTP)
SECTOR	Sektor / sub-sektor BEI, segmen usaha	Cat. umum, segmen	Threshold selection

2.2 Layer 2 — ratio dictionary (the 33 columns of input_2)

Grouped into seven families. "Src" shows where the formula is defined: a slide number in input_3, or *inferred* when the column exists in input_2 but input_3 gives no formula — the agent must fix those definitions explicitly in its config.

Family	Metric (column name)	Formula	Dir.	Src
Default model	Z-Score	$6,56 \cdot T1 + 3,26 \cdot T2 + 6,72 \cdot T3 + 1,05 \cdot T4 + 3,25$	↑	s.6,10
	$T1 \cdot T2$	Working Capital/TA · Saldo Laba/TA	↑	s.6,11
	$T3 \cdot T4$	EBITDA/TA · Total Ekuitas/Total Liabilitas	↑	s.6,7,11

Family	Metric (column name)	Formula	Dir.	Src
	Modified Z Score	$-3,337 + 0,736 \cdot WK_TA + 6,95 \cdot CASHPROF_TA + 0,864 \cdot SOLVR + 7,554 \cdot OPPROF_TA + 1,544 \cdot SALES_TA$	↑	s.12,13
Solvency / leverage	DER (X)	Total Liabilitas / Total Ekuitas (cek: $1/T4 = 3,23$ pada PPGD)	↓	s.7
	Debt Ratio (%)	Total Liabilitas / Total Aset	↓	s.7
	Long Term Debt To Equity (x) · ...To Total Asset (x)	Utang jangka panjang / Ekuitas · / Total Aset	↓	s.8
	Net Gearing Ratio (X)	(Utang jangka panjang – Kas) / Ekuitas (lihat isu §3)	↓	s.8
Liquidity	Current Ratio (x) · Quick Ratio (X)	$AL/LJP \cdot (AL - Persediaan)/LJP$	↑	s.7,8
	Piutang/Aset Lancar (%)	Piutang / Aset Lancar	↓	s.8
Coverage / debt service	Interest Coverage (x)	Laba Operasional / Beban Bunga	↑	s.7
	EBITDA/Interest Expense (X)	EBITDA / Beban Bunga	↑	inferred
	DSCR (x)	EBITDA / (Bunga + Pokok jatuh tempo)	↑	inferred
	Cash Flow Coverage Ratio (x)	Arus Kas Operasi / Beban Bunga (atau / utang)	↑	inferred
Profitability	Operating · Net Profit · EBITDA · Gross Margin (%)	$OI \cdot NI \cdot EBITDA \cdot (Rev - COGS) \div Pendapatan$	↑	s.9
	ROA (%) · ROE (%) · EBITDA/Total Aset (%)	$NI/TA \cdot NI/TE \cdot EBITDA/TA$	↑	inferred · s.8
Growth (YoY)	Sales · Operating Income · Net Income Growth (%)	(Periode ini – periode sama tahun lalu) / tahun lalu	↑	s.9 · inferred
Cash conversion	Operating Cash Flow/Net Income (X)	Arus Kas Operasi / Laba Bersih	↑	inferred

Identity columns: No., Kode (ticker), Nama Emiten, Tanggal Laporan. The DIFAPro filter adds four dimensions that the agent should carry as metadata: Periode, Tampilan Laporan (Tahunan/Semester/Kuartal), Sektor, Sub Sektor (slides 2–4).

2.3 Layer 3 — the rulebook (from input_3, as interpreted by the outputs)

Indicator	Threshold → category	Applies to
Altman Z-Score (EM)	$Z > 2,60 \rightarrow$ gagal bayar rendah (GREEN) · $1,21-2,60 \rightarrow$ meragukan (YELLOW) · $Z < 1,21 \rightarrow$ tinggi (RED)	Semua emiten
Modified Z-Score — model	$Z > 0 \rightarrow$ gagal bayar rendah · $Z \leq 0 \rightarrow$ gagal bayar tinggi	Semua emiten
Modified Z-Score — covenant band	> 0 Aman · $-0,5$ s.d. 0 Cukup Aman · ≥ -1 s.d. $-0,5$ Butuh Peninjauan · < -1 Tidak Aman	Non-finansial (di luar bank, multifinance, asuransi)
Interest Coverage (ICR)	Minimal 3x	Non-finansial
Current Ratio (CR)	Minimal 1,25x	Non-finansial
DER	Finansial $\leq 10x$ · Non-finansial infrastruktur $\leq 5x$ · Non-finansial non-infrastruktur $\leq 3x$	Per sektor
Indikator pendukung	Prospek sektor jangka panjang positif · Arus kas operasi positif 3 tahun terakhir (non-finansial) · Rating minimal A (stable) · Opini audit LK terakhir WTP	Semua; OCF hanya non-finansial
No threshold in materi	Debt Ratio, LTD ratios, Net Gearing, Quick, Piutang/AL, margins, growth, ROA/ROE, OCF/NI, DSCR ...	→ only "risk overlay" (HIGH/YELLOW), never "breach"

2.4 Layer 4 — context the user supplies

The Pegadaian outputs depend on information that is in none of the input files. The agent must therefore accept it as structured input, and label every number derived from it as an assumption:

- **Sector classification** — financial / non-financial infrastruktur / non-finansial non-infrastruktur. It switches thresholds on or off (Pegadaian = financial, so ICR 3x and CR 1,25x are *not* applied).

- **Risk overlay event** — e.g. "± Rp20 T exposure memerlukan realisasi jaminan, stress loss Rp2 T", plus recurrence scenarios (Rp2 T → Rp4 T → Rp6 T in 12 months).
- **Holdings** — per instrumen: nama seri, jenis (sukuk / obligasi), rating, kupon/imbal hasil, nilai nominal.
- **Two dates** — base date of the financials (30 Jun 2026) and assessment date (Sep 2026). The gap between them is what creates the "overlay".
- **Supporting facts** — current rating & outlook, latest audit opinion, sector outlook.

2.5 Layer 5 — outputs

Report A — issuer assessment (output 2) and **Report B — investor exposure memo** (output 1). Both are Bahasa Indonesia, narrative prose with one summary table, numbers in Indonesian format (koma desimal, titik ribuan, "Rp20 triliun"). Report B reuses Report A's conclusions verbatim and adds holdings, income math and question lists. Templates are in §4.3–4.4.

BAGIAN 3

Data-quality & methodology issues the agent must handle

Two kinds of problems appear in the samples. The agent should detect them automatically and report them, not silently use the numbers.

3.1 Anomalies in the ratio row (input_2, PT Pegadaian)

Observation	Why it is suspicious	Agent rule
Operating Income Growth = 88,05% and Piutang/Aset Lancar = 88,05%	Two unrelated ratios with identical value to 2 decimals → likely column-mapping error in export. Output 2 reports OI growth 88,05% as fact.	Flag any two different metrics that are equal; mark both "unverified" until recomputed.
Net Gearing = 2,86x and LTD/Equity = 2,86x	Formula subtracts cash; equality implies cash = 0 or the cash term was dropped.	Recompute from L1 if available; else note as unverified.
DSCR = 2,60x and EBITDA/Interest = 2,60x	DSCR should include principal; equality means principal was omitted.	Label DSCR "interest-only proxy".
Gross Margin = 100%	Financial company has no COGS line; the metric is meaningless, not excellent.	Suppress GM for financial sector.
T5 missing; MZ inputs (CASHPROF, SOLVR, OPPROF, SALES) not exported	Modified Z cannot be verified or stressed.	Require L1 data to compute and stress MZ.
Tanggal Laporan = 46203	Excel date serial, not text.	Convert serial → date (46203 = 30-06-2026).
Annualisation convention unknown	For a semester report, flow ratios change materially if not annualised. On DAAZ: MZ = 0,33 un-annualised vs 2,54 annualised.	Always state the convention; compute both when interim; decide on the annualised value and show the other as sensitivity.

3.2 Inconsistencies inside the rulebook (input_3)

Where	Inconsistency	Resolution used
Slide 6–7 (T2, T3, T4)	Descriptions are copied from the Modified Z table (slide 13): T2 says "laba bersih riil", formula is Saldo Laba/TA; T3 says "solvabilitas", formula is EBITDA/TA; T4 says "profitabilitas EBIT", formula is Ekuitas/Hutang.	Formulas win; slide 11 labels (Likuiditas, Kontinuitas, Profitabilitas, Solvabilitas) are used.
Slide 7 Current Ratio	Text says aset lancar excludes persediaan & piutang, formula is plain AL/LJP.	Use formula; that exclusion describes neither CR nor Quick.
Slide 8 Net Gearing	Text says interest-bearing debt; formula uses Utang Jangka Panjang — Kas.	Compute both; report slide formula as primary.

Where	Inconsistency	Resolution used
Slide 12 vs 14 (Modified Z)	Slide 12: $Z \leq 0$ = gagal bayar tinggi. Slide 14: $-0,5$ s.d. 0 = "Cukup Aman".	Report both lanes (model verdict + covenant band), as output 2 does.
Slide 14 call-outs	The "Minimal 3x" line is drawn closest to the Z-Score box; outputs read it as $ICR \geq 3x$.	Follow outputs ($ICR \geq 3x$, non-finansial); keep it configurable.
Slide 10 boundaries	Uses strict " $<$ " on both 1,21 and 2,60; exact values undefined.	Treat $1,21 \leq Z \leq 2,60$ as "meragukan".

BAGIAN 4

The analysis pattern

4.1 The reasoning chain behind both reports

Both Pegadaian reports follow the same ten moves, always in this order. This is the core "skill" the agent must reproduce:

1. **Fix the frame.** State base date, assessment date, sector assumption, and which thresholds therefore apply or do not apply.
2. **Score the default models.** Z-Score vs 1,21/2,60; Modified Z vs 0 and vs the covenant bands. Explain that the two scales differ, so 6,35 vs 0,33 is not a contradiction.
3. **Measure buffer, not only pass/fail.** "6,35 jauh di atas 2,60" vs "0,33 buffer tipis terhadap 0" — the thinner buffer gets the attention.
4. **Test applicable covenants** (DER, ICR, CR) with the sector-correct limit. A non-applicable limit is mentioned only as information, never as GREEN or breach.
5. **Scan non-threshold ratios** for concentrations and divergences (piutang/AL, quick ratio, OCF/NI negative) → "HIGH/YELLOW risk overlay".
6. **Apply the event overlay.** Convert the event into implied numbers (exposure Rp20 T, loss Rp2 T → recovery Rp18 T, recovery rate 90%, loss severity 10%). Call it a stress assumption, not an actual loss.
7. **Trace transmission channels:** asset quality → impairment/CKPN → laba → ekuitas → DER & Modified Z → liquidity/coverage → covenant headroom → debt service.
8. **Assign status in two lanes** and an overall status (rules in 4.2). Explain explicitly why it is not one level worse.
9. **List escalation triggers** (what would make it RED) and **data requests** (what is needed to finalise).
10. **Conclude** with base-case strength, the main risk, and the monitoring stance. Report B then maps all of this to the investor's holdings and turns every data gap into a question for the CFO or the Trustee.

4.2 Two-lane rating and the overall status

Lane	When used	Allowed labels	Example (PPGD)
Lane 1 — Covenant / model	Indicator has a numeric threshold in input_3 AND it applies to the sector	GREEN · YELLOW · RED (formal)	Z 6,35 GREEN · MZ 0,33 GREEN · DER 3,23x < 10x GREEN
Lane 2 — Risk overlay	No threshold in materi, threshold not applicable to sector, or forward-looking event	HIGH/YELLOW · YELLOW monitoring · GREEN (interpretasi arah)	Quick 0,12x HIGH/YELLOW · OCF/NI -3,74x HIGH/YELLOW · ICR 2,46x YELLOW (not breach, finansial)

Overall-status rules distilled from the outputs (the agent should implement these deterministically):

Overall	Condition (any one is enough unless stated)
RED	Modified $Z \leq 0$ on recomputed data · Z-Score $< 1,21$ · DER above sector limit · two or more applicable Lane-1 breaches · rating below A (stable) · payment default / Event of Default · actual loss materially above stress assumption and eroding equity
YELLOW - Enhanced Monitoring	One applicable Lane-1 breach · any Lane-2 HIGH item · a material overlay event not yet reflected in financials · a supporting indicator failed or unverified · Z in 1,21–2,60
GREEN	All applicable Lane-1 items GREEN, no Lane-2 HIGH, no open overlay, supporting indicators confirmed

Rule of thumb visible in both outputs: *never jump to RED on an assumption*. RED requires recomputed numbers that cross a model or covenant line; an un-booked stress loss can only push to YELLOW.

4.3 Report A template — "Analisis Menyeluruh Rasio Keuangan [Emiten]"

#	Section	Must contain
0	Judul + "Posisi Dasar [tgl] dengan Risk Overlay [bulan]" + Tujuan analisis	Base date, overlay date, reference to materi
–	Ringkasan Eksekutif (3–4 paragraf)	Model verdicts with numbers, overlay in one sentence with implied rates, overall status + "bukan RED" reasoning
1	Dasar Kriteria dan Asumsi Penilaian	Numbered list of every rule; sector assumption; explanation of the two lanes
2	Temuan Prioritas	The overlay/event first, with implied recovery & severity
3	Default Risk	Z & MZ, buffer comparison, "wajib dihitung ulang"
4–8	Kualitas Aset & Likuiditas · Cash Conversion · Leverage & Solvabilitas · Interest Coverage & Debt Service · Profitabilitas & Pertumbuhan	Each: numbers → threshold (or "tidak ada threshold") → status in bold
9	Analisis Stress	Bullet of assumptions; four impact channels; statement of what cannot be computed
10	Peta Risiko	Table: Prioritas · Area · Indikator · Interpretasi · Status (sorted by severity)
11–12	Posisi [bulan] · Trigger Eskalasi menjadi RED	Overall status; list of quantitative triggers
13–14	Data yang Harus Diminta · Kesimpulan Akhir	Data request list incl. rating & opini WTP; 3-paragraph conclusion
–	Sumber	Materi, file rasio, asumsi tambahan

4.4 Report B template — "Analisis Exposure [Investor] pada Surat Utang [Emiten]"

#	Section	Must contain
1	Ringkasan Eksekutif	Total nominal, split per instrument with %, base ratios, overlay, rolling stress (Rp2/4/6 T), status, transmission chain
2	Posisi Surat Utang	Table: Instrumen · Rating · Kupon · Nominal · Porsi; annual coupon = nominal × rate, labelled "indikatif"
3–5	Kondisi Dasar · Risk Overlay · Risiko Berulang	Re-uses Report A; recurring-loss scenario with "no double counting"
6–7	Implikasi terhadap Exposure · Posisi Risiko	Why loss on issuer ≠ loss on the bond; rating is not a substitute for monitoring
8	Pertanyaan kepada Direktur Keuangan	~7 questions: exposure breakdown, basis of loss, CKPN, stress impact, cash conversion, liquidity buffer, coverage sensitivity, breaking point
9	Pertanyaan kepada Trustee/Wali Amanat	~7 questions: per-instrument covenants, awareness of event, compliance calculation & headroom, reporting obligations, first trigger, EoD mechanics, actions
10–12	Pertanyaan Utama (waktu terbatas) · Kesimpulan · Basis Data & Batasan	One compound question in italics; conclusion; explicit limitations

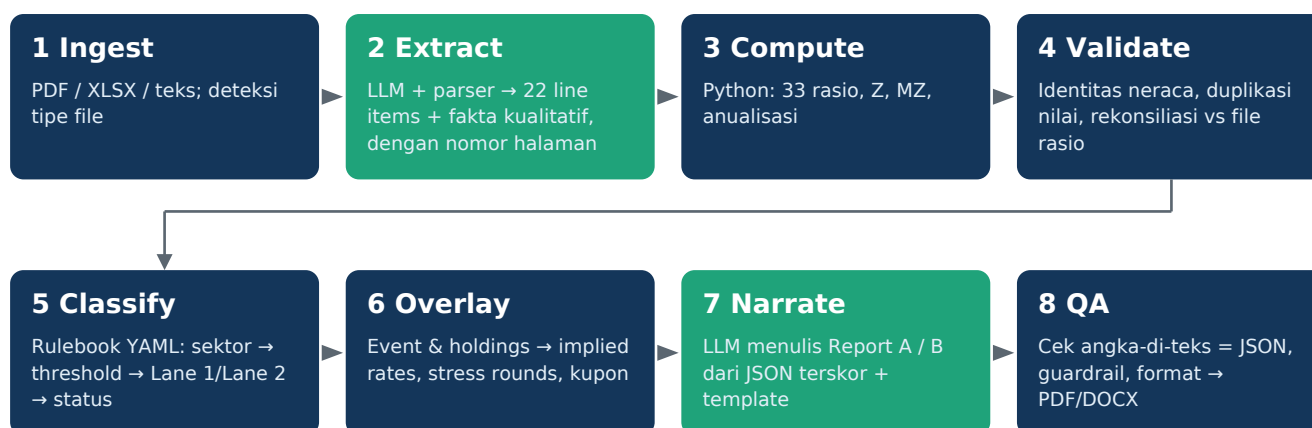
4.5 Writing guardrails (observed in every paragraph)

- Every status is followed by its threshold and source (" $< 10\times$ batas sektor finansial").
- Assumptions are named as such: "stress assumption, bukan loss aktual"; "perhitungan nominal sederhana, bukan proyeksi".
- Do not apply non-financial thresholds to financial issuers, and vice-versa — but do mention them as information.
- Never compute a pro-forma number without absolute inputs; say what is missing instead.
- No double counting of exposure or impairment across stress rounds.
- Treat each instrument separately (sukuk $\neq$ obligasi; terms must come from its own documents).
- Lead with the most material risk, not with the best-looking ratio.
- Indonesian number formatting, formal register, financial English terms allowed in italics-free form (exposure, recovery rate, headroom, covenant).

BAGIAN 5

AI agent blueprint

The key design decision: **numbers are computed by code, words are written by the LLM**. The outputs are long but formulaic; almost every sentence is a number, a threshold, a status, or a caveat. If the LLM only narrates a pre-scored JSON, it cannot invent or mis-round a ratio, and every status is reproducible.



■ LLM step ■ Deterministic code

5.1 What each component does

Component	Implementation	Notes
Extractor	LLM with a strict JSON schema over pdftotext layout text; table-aware for neraca/laba rugi; OCR fallback for scanned pages	Return value + source page + label as found. Reject if $TA \neq TL + TE$ (tolerance 0,1%).
Ratio engine	Pure Python, formulas from §2.2 config; annualisation factor from PERIOD (Semester $\times 2$, Kuartal $\times 4$)	Outputs both annualised and raw for interim; keeps formula version.
Validator	Rules from §3.1 (duplicate values, impossible margins, serial dates, identity checks); if both L1 and L2 present, recompute and diff	Every failed check becomes a "catatan data" line in the report.
Rulebook	Versioned YAML (thresholds, applicability by sector, label text)	Lets you change " $ICR \geq 3\times$ " without touching prompts.
Classifier	Deterministic Lane-1/Lane-2 assignment + overall rules from §4.2	Outputs status + reason codes + buffer to threshold.

Component	Implementation	Notes
Overlay module	Takes event text → structured scenario; computes recovery, severity, rolling rounds; holdings → porsi & kupon	All outputs tagged assumption=true.
Narrator	LLM, one call per section, given section template + only the JSON slice it needs	Low temperature; forbidden to output digits not present in its JSON slice.
QA	Regex every number in the draft and match it to JSON; guardrail checklist; Indonesian formatting	Fail → regenerate section with error list.

5.2 Input and output contracts

Input	Output (intermediate JSON)
<pre>{ "issuer": {"kode":"DAAZ","nama":"PT Daaz Bara Lestari Tbk", "sektor":"non_finansial","sub":"non_infras struktur"}, "period": {"base_date":"2026-06-30","type":"semester"}, "assessment_date":"2026-10-05", "sources": [{"type":"lk_pdf","path":"input_1.pdf"}, {"type":"ratio_xlsx","path":"input_2.xlsx"}], // optional "overlay": [{"desc":"Rp20 T exposure perlu realisasi jaminan", "exposure":2.0e13,"stress_loss":2.0e12," recurrence":[1,2,3]], // optional "holdings":[{"instrumen":"...Seri B","jenis":"obligas i","rating":"idA", "kupon":0.10,"nominal":3.0e9}], // optional -> Report B "supporting":{"rating":"idA","outlook":null,"audit_op inion":null,"sector_outlook":null}, "reports":["A","B"]} }</pre>	<pre>{ "line_items": {"TA":{"value":6616224661283,"page":"LK hal.1"}, ...}, "ratios": {"ICR":{"value":2.90,"annualised":null,"for mula":"OI/INT"}, ...}, "checks": [{"id":"DUP_VALUE","metrics":["OIG","AR_CA"], "severity":"warn"}], "status": {"ICR":{"lane":1,"limit":3.0,"status":"RED", "buffer":-0.10}, "overall":{"status":"YELLOW","reasons":["I CR<3x","CF0<0"]}}, "scenarios": [...], "report_md": "...", "report_pdf": "..."} }</pre>

5.3 Core system prompt for the Narrator (condensed)

Anda adalah analis kredit obligasi korporasi. Tulis bagian laporan "{section}" dalam Bahasa Indonesia formal berdasarkan HANYA data JSON berikut. Aturan:

- Setiap angka di teks harus berasal dari JSON; format Indonesia (koma desimal, "Rp2 triliun").
- Setiap status (GREEN/YELLOW/RED/HIGH) wajib disertai threshold dan sumber kriterianya.
- Bedakan "Lane 1 - covenant/model" (ada threshold yang berlaku untuk sektor) dan "Lane 2 - risk overlay" (tidak ada threshold / tidak berlaku / forward-looking). Lane 2 BUKAN pelanggaran.
- Threshold yang tidak berlaku untuk sektor disebut sebagai informasi, bukan dasar GREEN/RED.
- Asumsi (stress loss, skenario berulang, kupon nominal) selalu disebut asumsi/indikatif.
- Jangan hitung pro-forma bila nilai absolut tidak tersedia; sebutkan data yang dibutuhkan.
- Jelaskan mengapa status tidak satu tingkat lebih buruk (mis. "YELLOW, bukan RED, karena ...").
- Untuk tiap instrumen, perlakukan dokumen & covenant secara terpisah; jangan double counting.
- Urutkan dari risiko paling material. Prosa, tanpa bullet, kecuali template bagian meminta daftar.

5.4 Acceptance tests (use the Pegadaian pair as the golden set)

- Reproduction test:** feed input_2 + input_3 + the Rp20 T/Rp2 T overlay + YKPP holdings; the agent must reach YELLOW – Enhanced Monitoring, put asset quality first in the risk map, not apply ICR 3x / CR 1,25x, and compute Rp891,5 juta indicative annual return (Rp665 juta + Rp226,5 juta).
- Numeric fidelity:** 100% of numbers in text match JSON (QA regex), 0 invented figures.
- Rule coverage:** every applicable rule in §2.3 appears in "Dasar Kriteria"; every Lane-2 item says "bukan pelanggaran covenant".
- Anomaly detection:** the four duplicates/impossible values in §3.1 are flagged.
- Raw-statement test:** run on input_1 (DAAZ) and check balance identity and ratio values against an independent calculation (done in §6 and the Lampiran).

Analisis Menyeluruh Rasio Keuangan PT Daaz Bara Lestari Tbk

Posisi Dasar 30 Juni 2026 (Laporan Keuangan Interim, Tidak Diaudit) — Penilaian per Oktober 2026

Catatan: bagian ini adalah output yang dihasilkan dengan menjalankan pola di atas terhadap input_1. Semua angka dihitung ulang dari laporan keuangan (lihat Lampiran). Tidak ada data kepemilikan investor maupun event overlay yang diberikan untuk DAAZ, sehingga Report B hanya ditampilkan sebagai daftar pertanyaan dengan placeholder kepemilikan.

Tujuan analisis: menilai PT Daaz Bara Lestari Tbk (DAAZ) sebagai emiten obligasi menggunakan kriteria materi *Rasio Keuangan Obligasi / Modified Z-Score* (PT Edvisor Profina Visindo, 2026), berdasarkan angka absolut laporan keuangan konsolidasian interim 30 Juni 2026.

Ringkasan Eksekutif

Secara model, DAAZ berada pada profil **GREEN**. Z-Score sebesar **6,54** berada jauh di atas batas **2,60**, dan Modified Z-Score sebesar **2,54** (basis anualisasi) berada di atas **0** sehingga masuk kategori **Aman**. Dari sisi struktur modal, DER **1,96x** masih di bawah batas maksimum **3x** untuk emiten non-finansial non-infrastruktur, dan Current Ratio **1,57x** memenuhi batas minimum **1,25x**.

Namun, satu kriteria covenant yang berlaku untuk emiten non-finansial tidak terpenuhi: Interest Coverage Ratio (Laba Operasional/Beban Bunga) hanya **2,90x**, di bawah batas minimum **3x**, turun dari **4,07x** pada semester I 2025. Penurunan ini terjadi karena beban keuangan naik 30,4% sementara laba operasional turun 7,1%. Selain itu, arus kas operasi berbalik negatif **Rp-369,0 M** (semester I 2025: positif Rp427,8 M), terutama karena piutang usaha naik 35,4% dalam enam bulan sementara pendapatan hanya tumbuh 11,3% YoY.

Dengan demikian, posisi DAAZ per Oktober 2026 lebih tepat ditempatkan pada **YELLOW - Enhanced Monitoring**, bukan RED. Model default risk masih jauh dari batas, leverage dan likuiditas memenuhi covenant, dan emiten telah melunasi pokok Obligasi I Seri A Rp150 miliar pada 20 Juli 2026. Status YELLOW mencerminkan satu pelanggaran kriteria ICR berbasis data interim serta pelemahan cash conversion yang perlu dipantau sampai laporan tahunan 2026 tersedia.

1. Dasar Kriteria dan Asumsi Penilaian

DAAZ bergerak di perdagangan batu bara, nikel dan solar serta jasa angkutan laut, sehingga diperlakukan sebagai **sektor non-finansial non-infrastruktur**. Konsekuensinya, batas yang berlaku adalah: Z-Score ($Z > 2,60$ rendah; $1,21-2,60$ meragukan; $Z < 1,21$ tinggi), Modified Z-Score (> 0 Aman; $-0,5$ s.d. 0 Cukup Aman; ≥ -1 s.d. $-0,5$ Butuh Peninjauan; < -1 Tidak Aman), ICR minimal 3x, Current Ratio minimal 1,25x, DER maksimal 3x, serta indikator pendukung: prospek sektor positif, arus kas operasi positif tiga tahun terakhir, rating minimal A (stable), dan opini audit WTP.

Karena laporan bersifat semester, pos arus (pendapatan, laba, EBITDA, penyusutan) dianualisasi dengan faktor 2 untuk komponen Z-Score, Modified Z-Score, ROA dan ROE. Rasio yang merupakan perbandingan dua pos arus pada periode yang sama (ICR, margin, growth) tidak terpengaruh anualisasi. Hasil tanpa anualisasi ditampilkan sebagai sensitivitas.

2. Default Risk: GREEN

Z-Score sebesar **6,54** dibentuk oleh T1 (Modal Kerja/Total Aset) 0,209, T2 (Saldo Laba/Total Aset) 0,174, T3 (EBITDA/Total Aset) 0,122 dan T4 (Total Ekuitas/Total Liabilitas) 0,511. Tanpa anualisasi, nilainya tetap 6,13; keduanya jauh di atas 2,60, sehingga dikategorikan **GREEN**.

Modified Z-Score sebesar **2,54** pada basis anualisasi, dengan kontribusi terbesar dari SALES_TA (2,09) dan SOLVR (1,51). Tanpa anualisasi nilainya **0,33** — tetap di atas 0 dan masuk kategori Aman, tetapi dengan buffer yang tipis. Perbedaan besar antara dua basis ini menunjukkan bahwa Modified Z-Score DAAZ sangat sensitif terhadap perputaran aset (pendapatan/total aset), yang wajar untuk bisnis trading

bermargin tipis. Kesimpulan: **GREEN**, dengan catatan bahwa penurunan volume penjualan berdampak langsung pada indikator ini.

3. Interest Coverage dan Debt Service: RED pada kriteria ICR

ICR **2,90x** berada di bawah batas minimal **3x** yang berlaku untuk emiten non-finansial. Kekurangannya kecil: laba operasional semester I perlu sekitar **Rp10,8 M** lebih tinggi ($\pm 3,3\%$) untuk mencapai **3x**. EBITDA/Beban Bunga sebesar **3,62x** menunjukkan bahwa kemampuan tunai sebelum penyusutan masih menutup bunga dengan nyaman. Karena itu, indikator ICR dikategorikan **RED** sebagai temuan covenant, tetapi tidak cukup sendiri untuk menempatkan status keseluruhan pada RED.

DSCR indikatif (EBITDA anualisasi / (bunga anualisasi + pokok utang bank jatuh tempo 1 tahun + obligasi Seri A)) sebesar **1,37x**. Bila utang bank jangka pendek Rp795,8 M (fasilitas modal kerja revolving) ikut dihitung sebagai pokok, DSCR turun menjadi **0,58x**. Angka kedua menunjukkan ketergantungan pada perpanjangan (roll-over) fasilitas bank, sehingga status area ini **YELLOW** dan definisi pokok perlu dikonfirmasi sesuai perjanjian kredit.

4. Leverage dan Solvabilitas: GREEN

DER (Total Liabilitas/Total Ekuitas) **1,96x** berada di bawah batas **3x**. Debt Ratio **66,2%**, LTD/Ekuitas **0,87x**, LTD/Total Aset **0,29x**, dan Net Gearing **0,65x** (rumus materi) atau **1,16x** (seluruh utang berbunga dikurangi kas). Rasio utang berbunga terhadap ekuitas **1,39x** juga di bawah covenant bank BCA $\leq 4x$ (diuji pada laporan tahunan entitas anak). Kategori: **GREEN**.

5. Likuiditas dan Kualitas Aset Lancar: GREEN dengan catatan

Current Ratio **1,57x** memenuhi minimal **1,25x** walaupun turun dari **1,72x** pada 31 Desember 2025. Quick Ratio **1,23x** dan Piutang Usaha/Aset Lancar **45,5%** (tidak ada threshold di materi). Yang perlu dicatat adalah kas turun dari Rp841,9 M menjadi Rp494,6 M, sementara utang bank jangka pendek naik **34,5%** menjadi Rp795,8 M. Setelah pelunasan Seri A dan kupon ke-4 pada 20 Juli 2026, kas indikatif tersisa sekitar Rp341,2 M sebelum arus kas Juli. Status: covenant **GREEN**, risk overlay **YELLOW**.

6. Cash Conversion: HIGH/YELLOW

Arus kas operasi Rp-369,0 M menghasilkan OCF/Laba Bersih **-2,68x** dan Cash Flow Coverage (OCF/Beban Bunga) **-3,32x**. Pergeseran arus kas operasi sebesar Rp-796,8 M dibanding semester I 2025 sejalan dengan kenaikan piutang usaha Rp452,9 M dan pembayaran kepada pemasok yang melebihi penerimaan pelanggan. Indikator pendukung "arus kas operasi positif tiga tahun terakhir" berlaku untuk emiten non-finansial; untuk periode interim ini indikator tersebut tidak terpenuhi, sehingga perlu dikonfirmasi dengan arus kas tahunan 2023-2025 dan proyeksi 2026. Status: **HIGH/YELLOW**.

7. Profitabilitas dan Pertumbuhan: YELLOW

Pendapatan tumbuh **11,3% YoY**, tetapi laba operasional turun **7,1%** dan laba bersih turun **36,6%**. Gross Margin **6,38%**, Operating Profit Margin **4,68%**, EBITDA Margin **5,84%** dan Net Profit Margin **1,99%**. ROA **4,15%** dan ROE **12,29%** (anualisasi). Segmen jasa angkutan laut mencatat rugi sebelum pajak Rp18,1 miliar, dan kepentingan non-pengendali menanggung rugi Rp26,0 miliar, yang menandakan tekanan di sebagian entitas anak. Materi tidak memberikan threshold untuk rasio ini, sehingga status **YELLOW** merupakan interpretasi arah kinerja, bukan pelanggaran covenant.

8. Struktur Obligasi dan Peristiwa Setelah Periode Pelaporan

Instrumen	Nominal	Kupon	Tenor / jatuh tempo	Status
Obligasi I Daaz Bara Lestari 2025 Seri A	Rp150 miliar	8,85% tetap, triwulanan	370 hari	Lunas 20 Juli 2026 (pokok + kupon ke-4 Rp3,32 M)
Obligasi I Daaz Bara Lestari 2025 Seri B	Rp350 miliar	10,00% tetap, triwulanan	3 tahun sejak Juli 2025 ($\pm$ Juli 2028)	Beredar; bunga $\pm$ Rp35 M/tahun

Rating Pefindo **idA** (Single A) pada saat penawaran memenuhi batas minimal A, tetapi outlook "stable" dan rating terkini belum tercantum dalam laporan. Wali amanat adalah PT Bank Rakyat Indonesia (Persero) Tbk. Pelunasan Seri A tepat waktu merupakan bukti positif kemampuan pembayaran. Laporan interim ini tidak diaudit, sehingga opini WTP atas laporan tahunan terakhir perlu dikonfirmasi.

9. Peta Risiko Oktober 2026

Prio	Area	Indikator / Kondisi	Interpretasi	Status
1	Interest coverage	ICR 2,90x (S1 2025: 4,07x)	< 3x batas non-finansial; selisih $\pm$ Rp10,8 M laba operasional	RED (covenant)
2	Cash conversion	OCF Rp-369,0 M; OCF/NI -2,68x	Piutang naik 35,4%; indikator pendukung OCF positif tidak terpenuhi di periode ini	HIGH / YELLOW
3	Debt service / refinancing	DSCR 1,37x; 0,58x termasuk revolving	Ketergantungan roll-over utang bank jangka pendek	YELLOW
4	Profitabilitas	OI -7,1%; NI -36,6%; NPM 1,99%	Margin tipis; rugi segmen angkutan laut	YELLOW
5	Likuiditas	CR 1,57x; kas turun ke Rp494,6 M	$\geq 1,25x$; buffer kas menipis setelah pelunasan Seri A	GREEN / YELLOW
6	Modified Z-Score	2,54 (0,33 tanpa analisis)	> 0 Aman; sensitif terhadap perputaran aset	GREEN
7	Z-Score	6,54	> 2,60 kemungkinan gagal bayar rendah	GREEN
8	DER	1,96x	< 3x batas non-finansial non-infrastruktur	GREEN
9	Rating & audit	idA saat penerbitan; LK interim tidak diaudit	Outlook dan opini WTP terakhir perlu dikonfirmasi	CEK

10. Posisi Oktober 2026 dan Trigger Eskalasi menjadi RED

Overall assessment: YELLOW - Enhanced Monitoring. Status ini bukan pernyataan bahwa DAAZ memiliki kemungkinan gagal bayar tinggi; kedua model default risk masih GREEN dengan buffer lebar. Status RED perlu dipertimbangkan apabila salah satu kondisi berikut terbukti:

- ICR laporan tahunan 2026 tetap di bawah 3x, disertai pelanggaran kriteria lain (DER > 3x atau CR < 1,25x).
- Modified Z-Score hasil perhitungan ulang turun menjadi ≤ 0 , atau Z-Score turun ke bawah 1,21.
- Arus kas operasi tahunan 2026 negatif, sehingga indikator pendukung arus kas tiga tahun gagal.
- Rating turun di bawah A atau outlook menjadi negatif; opini audit bukan WTP.
- Pelanggaran covenant pinjaman bank BCA (CR $\geq 1x$, EBITDA/(bunga + pokok) $\geq 1x$, IBD/E $\leq 4x$ atau DER $\leq 2,5x$) yang berpotensi cross-default terhadap obligasi.

11. Data yang Harus Diminta

Laporan keuangan tahunan 2023-2025 (arus kas operasi tiga tahun), aging piutang usaha per pelanggan dan status penagihan pasca 30 Juni 2026, jadwal jatuh tempo dan rencana perpanjangan utang bank, perhitungan covenant BCA terakhir per entitas anak, rating Pefindo terkini beserta outlook, opini audit 2025, rincian penggunaan dana obligasi, serta proyeksi laba operasional semester II 2026.

12. Pertanyaan kepada Direktur Keuangan (Muljanto) dan Wali Amanat (PT BRI)

Elemen Report B. Data kepemilikan investor belum diberikan; tabel posisi akan berisi: Instrumen · Rating · Kupon · Nominal · Porsi, dengan kupon tahunan indikatif = nominal $\times$ 10,00% untuk Seri B.

Kepada Direktur Keuangan

1. Apa penyebab kenaikan piutang usaha pihak ketiga dari Rp1.228,8 miliar menjadi Rp1.722,3 miliar dalam enam bulan, berapa porsi yang sudah tertagih setelah 30 Juni 2026, dan apakah terdapat

konsentrasi pada pelanggan tertentu?

2. Bagaimana rencana manajemen mengembalikan ICR ke atas 3x, mengingat beban keuangan naik 30,4% sementara laba operasional turun 7,1%?
3. Berapa kas yang tersedia setelah pelunasan Seri A, dan bagaimana sumber dana untuk kupon Seri B serta pokok utang bank yang jatuh tempo 27 Desember 2026?
4. Apakah fasilitas modal kerja BCA akan diperpanjang, dan bagaimana hasil perhitungan covenant BCA per entitas anak saat ini?
5. Apa penyebab rugi segmen jasa angkutan laut dan rugi kepentingan non-pengendali, serta langkah perbaikannya?
6. Bagaimana sensitivitas laba operasional terhadap harga batu bara dan nikel, dan berapa titik di mana DER melampaui 3x atau ICR turun di bawah 2x?

Kepada Wali Amanat

1. Apa saja financial covenant dalam Perjanjian Perwaliamanatan Obligasi I Seri B, berapa nilai aktual dan headroom terakhir, dan kapan pengujian berikutnya?
2. Apakah perjanjian memuat klausul cross-default terhadap pinjaman bank, dan apakah Wali Amanat menerima laporan kepatuhan covenant BCA?
3. Apakah penurunan ICR atau arus kas operasi negatif termasuk kejadian yang wajib dilaporkan Emiten, dan apakah kewajiban tersebut telah dipenuhi?
4. Apakah Wali Amanat telah menerima konfirmasi pembayaran pokok dan kupon Seri A tanggal 20 Juli 2026 serta kupon Seri B yang terakhir jatuh tempo?
5. Tindakan apa yang akan diambil bila rating turun di bawah A atau terjadi pelanggaran covenant, termasuk mekanisme pemberitahuan, masa perbaikan, dan RUPO?

Pertanyaan utama apabila waktu terbatas: Dengan ICR 2,90x di bawah batas 3x, arus kas operasi negatif Rp369 miliar, dan kenaikan piutang Rp452,9 miliar dalam enam bulan, mohon Direktur Keuangan menjelaskan rencana penagihan dan sumber dana untuk kupon serta pokok Seri B dan utang bank yang jatuh tempo; dan mohon Wali Amanat mengonfirmasi headroom covenant Seri B serta ada tidaknya risiko cross-default dari fasilitas BCA.

13. Kesimpulan

Berdasarkan laporan keuangan interim 30 Juni 2026, DAAZ masih berada pada posisi kuat dalam model default risk (Z-Score 6,54, Modified Z-Score 2,54) dan memenuhi covenant DER serta Current Ratio. Pelunasan Seri A tepat waktu memperkuat bukti kemampuan bayar.

Namun, ICR yang turun di bawah 3x, arus kas operasi negatif akibat penumpukan piutang, penurunan laba bersih 36,6% dan ketergantungan pada fasilitas bank jangka pendek menempatkan DAAZ pada **YELLOW - Enhanced Monitoring**. Penilaian ini bersifat sementara karena berbasis data interim tidak diaudit; status dapat kembali GREEN bila laporan tahunan 2026 menunjukkan ICR $\geq$ 3x dan arus kas operasi positif, atau naik ke RED bila trigger pada bagian 10 terpenuhi.

Sumber kriteria: Materi Rasio Keuangan Obligasi / Modified Z-Score, PT Edvisor Profina Visindo, 2026 (slide 6-14).

Sumber data: Laporan Keuangan Konsolidasian Interim PT Daaz Bara Lestari Tbk 30 Juni 2026 (tidak diaudit), termasuk Catatan 9, 11, 13, 29 dan 32.

LAMPIRAN

Ekstraksi line item & jejak perhitungan (DAAZ)

Seluruh angka dalam Rupiah penuh, diambil dari input_1. "Hal." mengacu pada nomor halaman laporan keuangan (bukan halaman PDF). Identitas neraca terverifikasi: Total Liabilitas + Total Ekuitas = Total Aset = Rp6.616.224.661.283.

Field	Nilai 30 Jun 2026 / S1 2026	Pembandingan	Sumber
Total Aset	6.616.224.661.283	6.318.716.272.021	Hal. 1
Aset Lancar	3.809.837.654.928	3.645.243.258.216	Hal. 1
Liabilitas Jangka Pendek	2.426.953.861.957	2.120.404.160.530	Hal. 1
Total Liabilitas	4.378.915.557.304	4.140.972.615.176	Hal. 1
Total Ekuitas	2.237.309.103.979	2.177.743.656.845	Hal. 2
Saldo Laba (total)	1.150.142.452.031	-	Hal. 2
Persediaan	819.172.223.768	-	Hal. 1
Piutang Usaha	1.733.023.520.048	1.280.142.676.044	Hal. 1
Kas dan Bank	494.565.942.864	841.923.738.835	Hal. 1
Utang Bank Jangka Pendek	795.832.844.373	591.870.249.907	Hal. 1
Utang Bank JT 1 Tahun	214.206.151.243	-	Hal. 1
Utang Obligasi (lancar)	150.000.000.000	-	Hal. 1; Cat. 13
Utang Bank Jangka Panjang	1.590.991.038.098	-	Hal. 1
Utang Obligasi (jk. panjang)	350.000.000.000	-	Hal. 1; Cat. 13
Pendapatan	6.908.103.651.164	6.205.134.617.775	Hal. 3
Beban Pokok Pendapatan	6.467.047.314.121	-	Hal. 3
Laba Kotor	441.056.337.043	443.079.644.379	Hal. 3
Beban Usaha	117.890.563.936	95.354.277.500	Hal. 3
Laba Operasional (GP – Beban Usaha)	323.165.773.107	347.725.366.879	dihitung
Beban Keuangan	111.317.622.205	85.383.418.762	Hal. 3
Laba Bersih	137.448.447.134	216.789.101.049	Hal. 3
Penyusutan	80.097.531.868	-	Hal. 50 (Cat. 9)
EBITDA (OI + D&A;)	403.263.304.975	-	dihitung
Arus Kas Operasi	(369.028.883.418)	427.762.048.988	Hal. 5
Modal Kerja (CA – CL)	1.382.883.792.971	-	dihitung
Utang Berbunga (IBD)	3.101.030.033.714	-	dihitung
Utang Jangka Panjang (LTD)	1.940.991.038.098	-	dihitung

Pembandingan: posisi 31 Des 2025 untuk pos neraca; semester I 2025 untuk pos laba rugi dan arus kas.

Hasil rasio (33 kolom skema input_2)

Metrik	Nilai	Catatan	Metrik	Nilai	Catatan
Z-Score	6,54		Net Gearing Ratio (X)	0,65	rumus materi; 1,17 basis IBD
Modified Z Score	2,54	anualisasi; 0,33 tanpa	Sales Growth (%)	11,33	YoY
T1	0,21		Operating Income Growth (%)	-7,06	YoY
T2	0,17		Net Income Growth (%)	-36,60	YoY
T3	0,12	anualisasi	Operating Profit Margin (%)	4,68	
T4	0,51		Net Profit Margin (%)	1,99	
DER (X)	1,96		EBITDA Margin (%)	5,84	
Interest Coverage (x)	2,90		Gross Margin (%)	6,38	
Current Ratio (x)	1,57		EBITDA/Interest Expense (X)	3,62	
Debt Ratio (%)	66,18		ROA (%)	4,15	anualisasi
LTD To Equity (x)	0,87		ROE (%)	12,29	anualisasi
LTD To Total Asset (x)	0,29		Cash Flow Coverage Ratio (x)	-3,32	OCF/Bunga
Quick Ratio (X)	1,23		DSCR (x)	1,37	0,58 termasuk revolving
EBITDA/Total Aset (%)	12,19	anualisasi	Operating Cash Flow/Net Income (X)	-2,68	
Piutang/Aset Lancar (%)	45,49	piutang usaha			

Jejak Modified Z-Score (anualisasi)

$Z = -3,337 + 0,736 \times 0,2090 + 6,95 \times 0,0658 + 0,864 \times 1,5109 + 7,554 \times 0,0977 + 1,544 \times 2,0882 = 2,541$. CASHPROF = (Laba Bersih + Penyusutan) $\times$ 2; OPPROF = Laba Operasional $\times$ 2; SALES = Pendapatan $\times$ 2; SOLVR = Total Aset / Total Liabilitas. Amortisasi tidak diungkapkan terpisah dan diasumsikan nol.

Non-Controlling Investment Monitoring Agent

Auto-drafting YKPP's Nota Dinas from PT Pesonna Optima Jasa's monthly financial statements

Version	1.0 — Draft for review
Date	5 Oktober 2026
Owner	Divisi Keuangan & Subsidiary, YKPP
Readers	Analyst, Kepala Divisi, engineering team building the agent
Reference samples	Input: Laporan Keuangan PT POJ periode Juli 2026 (PDF, 55 hlm). Output: Nota Dinas Analisis Kinerja PT POJ periode Agustus 2026 (DOCX).
Amounts	Source statements are in ribuan Rupiah . In this PRD, "M" = miliar Rupiah, "T" = triliun Rupiah.

1. Summary

Every month PT Pesonna Optima Jasa (PT POJ) sends its parent a financial package: statement of financial position, profit or loss, cash flow, notes, and an RKAP achievement report. YKPP holds 1.000 of PT POJ's 100.000 shares (1%) and turns that package into a **Nota Dinas** for Pengurus YKPP. The memo re-reads the subsidiary's numbers from a non-controlling shareholder's point of view, sorts findings into **Merah / Kuning / Hijau**, and attaches critical questions to be raised through shareholder channels.

The work is manual but highly patterned. Comparing the two samples shows that roughly four-fifths of the memo is either fixed template text or deterministic arithmetic on about 40 source fields; the remaining fifth is judgement-shaped prose that follows a repeatable four-beat structure. This PRD specifies an agent that extracts, validates, computes, classifies and drafts, so the analyst only reviews, adjusts and signs.

Core design principle — the LLM writes, code calculates. Every number that appears in the memo must be a pre-computed, pre-formatted value with a source page reference. The language model is never allowed to do arithmetic or introduce a figure that is not in the metric file.

2. Background and problem

2.1 Parties

Entity	Role	Why it matters to the agent
PT Pesonna Optima Jasa (PT POJ)	Reporting entity; outsourcing, rental, expedition, used-vehicle sales	Source of all input data
PT Pegadaian Galeri Dua Empat	Controlling shareholder, 99.000 shares (99%)	Addressee of the input package; controls dividends and corporate actions
YKPP (Yayasan Kesejahteraan)	Non-controlling shareholder, 1.000 shares (1%)	Author of the output; every finding is framed around YKPP's economic interest
Pemerintah Republik Indonesia	Ultimate parent	Context only

2.2 Problem

Preparing the memo means reading a 55-page scanned package, re-keying figures, computing ratios, and writing a governance-sensitive narrative. The samples already show three failure modes that an agent must prevent, not repeat:

- **Period mismatch.** The August memo's comparatives (revenue Rp1,062 T, laba Rp60,51 M, utang usaha Rp8,20 M, beban keuangan Rp2,46 M) are the *July 2025 YTD* figures. Comparing 8 months of 2026 with 7 months of 2025 inflates growth: the memo reports +42,4% revenue growth, while July-on-July growth is +27,0%.
- **Unreliable text layer.** The PDF's embedded text interleaves columns. Naive parsing reads total current liabilities as Rp363,94 M instead of Rp653,35 M, which turns a current ratio of 0,60x into 1,07x and would move the top Merah finding to green.
- **Inconsistent source.** The July package contains bilingual note contradictions (Notes 6 and 7), an interest payment shown with the wrong sign, a cash-flow total labelled "penurunan" on an increase, and a promised equity statement and RKAP section that are absent.

3. The input → output pattern

3.1 Input taxonomy

The package has nine layers. Only some of them feed the memo; the extractor should prioritise accordingly.

Layer	Pages (Jul-26)	Content	Priority
A. Cover letter	2	Letter number, date, period, list of attachments	Metadata
B. Statement of financial position	5–6	9 current-asset lines, fixed assets; 9 current + 5 non-current liability lines; equity (share capital, OCI remeasurement, retained earnings appropriated / unappropriated)	P0
C. Profit or loss — YTD	7	Revenue → COGS → gross profit → opex → finance cost → other → PBT → tax → net profit → OCI	P0
D. Profit or loss — single month	8	Same lines for the reporting month only	P1
E. Cash flow (direct method)	9	Receipts, interest, suppliers, employees → CFO; capex, asset sales → CFI; borrowings, repayments, dividends → CFF; opening / closing cash	P0
F. Notes 1–3	10–45	Corporate info, governance, accounting policies, estimates	Low (Note 1 governance only)

Layer	Pages (Jul-26)	Content	Priority
G. Notes 4–24	46–53	Breakdowns: bank list (4), AR gross / unidentified / ECL (5), prepaid (8), other assets (9), fixed-asset roll-forward (11), taxes (12), AP by type (13), accruals incl. dividend payable (14), borrowings ST / LT (16), revenue and COGS by segment (20–21), opex (22), ownership (19)	P0 for 5, 11–16, 19–22
H. RKAP achievement	absent	Target vs actual per line, ±5% proportional variance, management explanation (cover letter item 6)	P0 — separate required input
I. UMP annex	54–55	118 cash-advance lines: number, date, due date, age, PIC, purpose, amount. Total Rp2.495.823 ribu ties to Note 9.	P1

3.2 Output taxonomy (Nota Dinas skeleton)

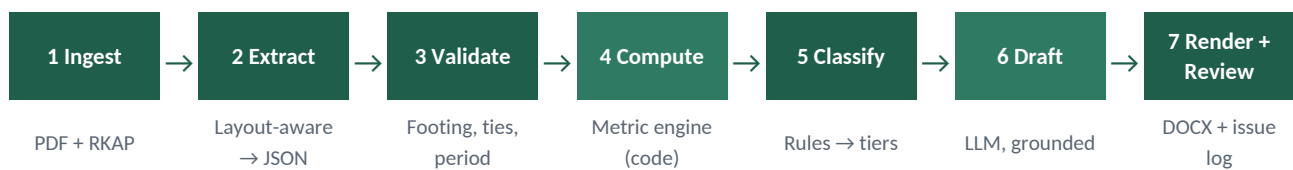
#	Block	Type	Rule
1	Title + header table	Fixed + variables	NOTA DINAS / YAYASAN KESEJAHTERAAN PEGADAIAN PERMATA. Nomor and Tanggal left as dotted blanks. Kepada: Pengurus YKPP. Dari: Divisi Keuangan & Subsidiary. Perihal: Analisis Kinerja Keuangan dan Monitoring Investasi YKPP pada PT Pesonna Optima Jasa Periode {Bulan Tahun}.
2	Purpose paragraph	Fixed + period	Ownership context and that the FS for the period is available.
3	Non-controlling positioning	Fixed	Monitoring does not enter management or operational control; open items become clarification via shareholder rights.
4	Headline	Template + metrics	Revenue YTD, growth %, comparative; net profit, growth %.
5	Holistic pivot	Fixed	Growth must be read together with financial position; lists YKPP concerns.
6	Tier legend	Fixed	Findings grouped Merah, Kuning, Hijau; Merah first.
7	I. MERAH	Generated	1–n numbered items, each in the four-beat format (3.3).
8	II. KUNING	Generated	One paragraph per theme, no questions, hedged language.
9	III. HIJAU	Generated	Positives, always conditional on cash conversion and shareholder return.
10	IV. KESIMPULAN DAN PENUTUP	Template	Restates headline, lists Merah themes, repeats governance disclaimer.
11	Closing line	Fixed	Bold: Demikian Nota Dinas ini disampaikan...

3.3 Anatomy of a Merah item

Every Merah item in the sample follows the same four beats. This is the most formulaic part of the memo and the easiest to generate reliably.

Beat	Purpose	Pattern from the August memo
1. Heading	Name the topic; put the metric in the title when it is striking	3. <i>Piutang Usaha Pihak Ketiga Mencapai 301,20% terhadap RKAP</i>
2. Fact paragraph	Numbers, comparison, and management's own explanation if the source gives one	<i>...sekitar Rp91,18 miliar, dibandingkan target... sekitar Rp30,27 miliar... Dalam laporan dijelaskan bahwa...</i>
3. Relevance to YKPP	Translate into shareholder impact: dividends, refinancing, capital call, quality of earnings, value of the stake	<i>...dapat memengaruhi kualitas laba dan kebutuhan modal kerja.</i>
4. Pertanyaan kritis	One bold sentence, 3–4 questions joined by semicolons, last one introduced by "serta"	<i>Pertanyaan kritis: bagaimana aging piutang...; berapa bagian yang telah tertagih...; serta apa yang menyebabkan...</i>

3.4 Transformation pipeline



Stages 1–5 and 7 are deterministic software. Only stage 2 (vision extraction) and stage 6 (drafting) use a language model, and both are wrapped by deterministic checks: extraction by the validation rules in section 7, drafting by the numeric grounding check in FR-07.

3.5 Writing style rules

Rule	Specification	Example
Hedge	"sekitar" before every amount and most ratios	sekitar Rp99,87 miliar
Scale	Source is in ribuan. value ÷ 1.000.000 = miliar. Below Rp1.000 M → miliar, 2 decimals; at or above → triliun, 3 decimals	Rp84,79 miliar; Rp1,512 triliun
Number format	Comma decimal, dot thousands	0,52x; 2,19x; 301,20%
Precision	Growth 1 decimal; margins and RKAP achievement 2 decimals; ratios 2 decimals + "x"	42,4%; 10,10%; 176,88%
Naming	First mention "PT Pesonna Optima Jasa (PT POJ)", then "PT POJ"; YKPP never expanded in body	—
Vocabulary	English finance terms kept in-line, as in the sample	negative working capital, refinancing, maturity profile, covenant, accrued income
Stance	Never instructs PT POJ management. Uses "perlu dicermati", "perlu klarifikasi", "bagi YKPP"	—
Positives	Always conditional	faktor positif sepanjang...
Layout	Arial 10,5 pt body, 14 pt bold title, justified, US Letter, margins 2,5 cm L/R and 2 cm T/B; tiers named in text, no colour	Taken from the sample DOCX

4. Goals and non-goals

Goals

ID	Goal	Measure
G1	A reviewable draft exists shortly after the package arrives	Draft ready ≤ 15 minutes after PDF + RKAP upload
G2	Every figure is traceable	100% of numbers in the draft link to a source page and field
G3	Classification matches the analyst	≥ 90% tier agreement over 3 shadow cycles
G4	Analyst effort drops	Review and edit ≤ 30 minutes per memo (baseline to be measured, see Q6)
G5	Source errors are caught before they reach Pengurus	100% detection on the seeded-error fixture set

Non-goals (v1)

- **No sending or signing.** The agent never issues the Nota Dinas; Nomor and Tanggal stay blank for the secretariat.
- **No management instructions.** Output stays within non-controlling shareholder rights; no operational recommendations.
- **No valuation model.** Fair value of YKPP's stake (DCF, multiples) is a separate initiative.
- **Single subsidiary.** Schema is built for PT POJ; other investees are P2.
- **No direct correspondence with PT POJ.** Drafting clarification letters is P2.

5. Users and stories

Persona	Story
Analyst, Divisi Keuangan & Subsidiary	As the analyst, I want to upload the monthly package and RKAP and receive a draft memo with an evidence table, so that I spend my time checking judgement rather than re-keying numbers.
Analyst	As the analyst, I want the agent to stop and tell me when periods do not match or totals do not foot, so that I never publish a growth rate built on mismatched data.
Analyst	As the analyst, I want to override a tier or edit a question and have the change recorded, so that thresholds can be recalibrated from my decisions.
Kepala Divisi	As the reviewer, I want a one-page issue log listing source inconsistencies and missing inputs, so that I know what the memo could not assess.
Pengurus YKPP	As a board member, I want the memo in the familiar format with Merah items first, so that I can see in two minutes what needs clarification from PT POJ.
Analyst (edge case)	As the analyst, when the RKAP sheet is missing I want the draft to be produced anyway with RKAP sentences clearly marked as unavailable, so that the rest of the analysis is not blocked.

6. Functional requirements

P0 — must have

ID	Requirement	Acceptance criteria
FR-01	Ingest the PT POJ PDF (scanned, signed pages included) and the RKAP file (XLSX or PDF).	Detects reporting month and year from statement headers. Refuses to proceed if the period cannot be determined.
FR-02	Layout-aware extraction into the data contract (section 9). Vision or table extraction; the raw PDF text layer must never be the sole source.	On the July 2026 fixture, every P0 field equals the reference values in section 8. Each field stores page number.
FR-03	Validation engine applying rules V-01 to V-10 (section 7) before any computation.	Blocking failures stop the run with a readable message; warnings flow into the issue log.
FR-04	Deterministic metric engine for catalogue M-01 to M-16, outputting raw and Indonesian-formatted values.	Unit tests reproduce the Jul-26 column of section 8 exactly.
FR-05	Rules engine assigning Merah / Kuning / Hijau per signal, thresholds in a config file.	Each signal outputs tier, triggering value and rule ID. Changing a threshold needs no code change.
FR-06	Question bank per Merah topic (section 11).	Writer selects 3–4 questions per item, adapted with figures.
FR-07	Grounded drafting by the LLM, using the template (appendix A), metrics, tiers and selected questions.	Post-check: every number in the draft matches a metric value; any unmatched number fails the draft. No topic appears that is not in the classification output.
FR-08	Render DOCX matching the style rules in 3.5.	Header table, bold section titles and bold Pertanyaan kritis, justified text.
FR-09	Review package: draft, issue log, evidence table (figure → field → page).	Analyst can trace every number in under one click or lookup.

P1 — next

ID	Requirement	Why
FR-10	Hak Ekonomi YKPP module: dividend declared, paid and payable; YKPP's 1% share; dividend payable vs cash.	July shows Rp110,27 M dividend payable (Note 14) against Rp43,08 M cash; YKPP's share ≈ Rp1,10 M. The most direct shareholder signal in the package, not covered by the current template.
FR-11	Cash quality: CFO / net profit, free cash flow (CFO – capex).	July CFO / laba = 1,50x (good) but capex Rp216,34 M > CFO Rp117,70 M; FCF –Rp98,64 M.
FR-12	Receivable quality: ECL coverage, unidentified receipts.	ECL coverage fell from 6,27% to 3,33% while gross AR rose 35,3%.
FR-13	UMP aging from the annex: overdue advances by PIC and age.	Several advances are 23–75 days old.
FR-14	Month-over-month trend using prior memos and the single-month P&L.	Shows direction, not just YTD level.

P2 — later

- Other subsidiaries and investees on the same schema.
- Monitoring dashboard across months.
- Draft clarification letter to PT POJ generated from Merah questions.
- Year-end mode using audited statements.

7. Validation rules

ID	Rule	Severity	Seen in samples
V-01	Footing: each subtotal equals the sum of its lines (± 1 ribu rounding)	Blocking	Catches text-layer misreads
V-02	Total assets = total liabilities + total equity	Blocking	Holds in July (1.487.664.962)
V-03	Face-to-note ties: BS Utang Lainnya = Note 14 Utang Lainnya + Utang Dividen; BS borrowings = Note 16; BS fixed assets = Note 11 book value; Note 9 UMP = annex total	Warning	112.433.909 = 2.166.018 + 110.267.891; UMP 2.495.823 ties
V-04	Period alignment: current and comparative YTD cover the same number of months	Blocking for growth metrics	August memo compared Aug-26 YTD with Jul-25 YTD
V-05	Cash flow closing cash = BS Kas dan Bank	Blocking	Holds in July (43.079.320)
V-06	Bilingual consistency of amounts in note prose; face statement wins	Warning	Note 6 ID 14.266.409 vs EN 4.971.487; Note 7 ID 101.922.890 vs EN / BS 97.007.537
V-07	Sign and label sanity in the cash flow	Warning	Interest paid shown as +145.731; "Penurunan bersih" on +3.701.738
V-08	Required inputs present: RKAP section, equity statement	Warning	Both absent in July, though listed in the cover letter and contents
V-09	Plausibility: implied annual borrowing cost = finance cost YTD $\times 12$ / months $\div$ borrowings within 3–15%	Warning	July: 0,48% p.a. on Rp654,44 M
V-10	Date sanity: completion and authorisation dates consistent with the period	Info	Note 26 says completed 2 July 2026; Note 1 cites 31 May 2026

8. Metric catalogue and July 2026 reference fixture

The Jul-26 and Jul-25 columns were computed from the July package and verified against page images. They are the expected values for the extraction and metric unit tests. Aug-26 values are what the sample memo reported.

ID	Metric	Formula	Jul-26	Jul-25	Aug-26 memo	Used in
M-01	Pendapatan YTD	P&L	Rp1,348 T	Rp1,062 T	Rp1,512 T	Headline, Hijau
M-02	Revenue growth	Rev $\div$ Rev pembanding – 1	27,0%	—	42,4%*	Headline
M-03	Laba tahun berjalan	P&L	Rp78,44 M	Rp60,51 M	Rp87,65 M	Headline, Hijau
M-04	Net profit growth	NI $\div$ NI pembanding – 1	29,6%	—	44,8%*	Headline
M-05	Gross margin	GP $\div$ Rev	10,04%	10,47%	10,10%	Kuning
M-06	Net margin	NI $\div$ Rev	5,82%	5,70%	5,80%	Hijau
M-07	Current ratio	CA $\div$ CL	0,60x	0,67x	0,52x	Merah 1
M-08	Working capital	CA – CL	–Rp263,47 M	–Rp152,84 M	–Rp347,18 M	Merah 1
M-09	Liabilities / equity	TL $\div$ TE	2,16x	1,89x	2,19x	Merah 2
M-10	Total pinjaman	Note 16 ST + LT	Rp654,44 M	Rp654,22 M	Rp727,97 M	Merah 2
M-11	Accrued income growth	Δ balance	46,0%	—	$\approx 50\%$	Merah 4
M-12	Utang usaha	Balance, Δ	Rp30,97 M (+277,5%)	Rp8,20 M	Rp84,79 M	Merah 5

ID	Metric	Formula	Jul-26	Jul-25	Aug-26 memo	Used in
M-13	Utang pajak	Balance, Δ	Rp63,46 M (+51,2%)	Rp41,98 M	Rp66,40 M	Merah 6
M-14	Fixed assets / total assets	FA ÷ TA	73,8%	—	largest asset	Kuning
M-15	Implied borrowing cost	FC × 12/m ÷ debt	0,48% p.a.	—	flagged	Kuning
M-16	RKAP achievement per line	Actual ÷ target	not available	—	301,20%; 176,88%...	Merah 3, Kuning

* Inflated by comparing 8 months with 7 months (V-04).

9. Data contract (extraction schema, abridged)

```
{
  "meta": { "entity": "PT POJ", "period_end": "2026-07-31", "ytd_months": 7,
    "comparative_end": "2025-07-31", "comparative_months": 7, "unit": "ribu_rupiah" },
  "bs": { "kas_bank": { "cur": 43079320, "cmp": 20465755, "page": 5 },
    "piutang_usaha_bersih": { ... }, "pendapatan_ymh_diterima": { ... },
    "total_aset_lancar": { ... }, "aset_tetap_bersih": { ... },
    "utang_usaha": { ... }, "utang_pajak": { ... }, "pinjaman_lancar": { ... },
    "utang_lainnya": { ... }, "total_liabilitas_lancar": { "cur": 653348363, ... },
    "pinjaman_tidak_lancar": { ... }, "kewajiban_pihak_berelasi": { ... },
    "total_liabilitas": { ... }, "total_ekuitas": { ... } },
  "pl_ytd": { "pendapatan": { ... }, "beban_pokok": { ... }, "laba_bruto": { ... },
    "beban_usaha": { ... }, "beban_keuangan": { ... }, "laba_bersih": { ... } },
  "pl_month": { ... same keys ... },
  "cf": { "cfo": { ... }, "capex": { ... }, "penerimaan_pinjaman": { ... },
    "pembayaran_pinjaman": { ... }, "dividen_dibayar": { ... }, "kas_akhir": { ... } },
  "notes": { "n5": { "bruto": ..., "belum_teridentifikasi": ..., "ckpn": ... },
    "n13": { "aset_tetap": ..., "asuransi": ..., "perbaikan": ..., "lainnya": ... },
    "n14": { "utang_dividen": 110267891, ... }, "n16": { "st": ..., "lt": ... },
    "n19": { "ykpp_shares": 1000, "total_shares": 100000 } },
  "rkap": [ { "line": "piutang_pihak_ketiga", "target": ..., "actual": ..., "explanation": "..." } ],
  "ump": [ { "no": "UMP/26/0003900", "age_days": 75, "pic": "...", "amount": 13600 } ],
  "issues": [ { "rule": "V-06", "severity": "warning", "message": "...", "page": 47 } ]
}
```

10. Classification rules (initial calibration)

Thresholds are inferred from a single output sample and must be calibrated against at least six analyst-labelled months (Q4). Applied to the August figures, these rules reproduce the sample memo's tiering: six Merah items, four Kuning themes, and a Hijau section.

Signal	MERAH	KUNING	HIJAU	Jul-26 result
Liquidity	CR < 1,0x or NWC < 0	CR 1,0–1,2x	CR > 1,2x	MERAH (0,60x)
Leverage (TL / TE)	> 2,0x or +0,2x YoY	1,5–2,0x	< 1,5x	MERAH (2,16x)
RKAP risk lines (AR, tax payable, AP)	≥ 150% of target	Outside ±5% band	Within ±5%	n/a (no RKAP)
RKAP other lines (cash, other assets, FA)	—	Outside ±5% band	Within ±5%	n/a (no RKAP)
Accrued income	Growth > revenue growth	0 < growth ≤ revenue growth	Decline	MERAH (46,0% vs 27,0%)
Trade payables	Growth > 100%	25–100%	< 25%	MERAH (+277,5%)

Signal	MERAH	KUNING	HIJAU	Jul-26 result
Tax payable	≥ 150% of RKAP, or > 50% YoY	20–50% YoY	< 20%	MERAH (+51,2%)
Gross margin	Down > 1,0 pp	Down 0–1,0 pp	Flat or up	KUNING (–0,43 pp)
Finance cost vs borrowings	—	Directions diverge, or V-09 fails	Consistent	KUNING (0,48% p.a.)
Asset intensity (FA / TA)	—	> 60% (standing monitor)	—	KUNING (73,8%)
Revenue and net profit growth	Both negative	One negative	Both positive	HIJAU
Net margin	Down > 1,0 pp	Down ≤ 1,0 pp	Up	HIJAU (5,70% → 5,82%)

Merah items are ordered: Likuiditas, Leverage, Piutang, Accrued income, Utang usaha, Kewajiban perpajakan, then any new topic. When RKAP is missing, the RKAP-based signals fall back to year-on-year rules and the issue log says so.

11. Question bank (Merah topics)

Questions are written in the memo's register. The writer picks 3–4 per item and inserts the relevant figures and dates.

Topic	Candidate questions
Likuiditas & negative working capital	kecukupan likuiditas 12 bulan ke depan; maturity profile kewajiban; kebutuhan refinancing material; potensi kebutuhan tambahan modal atau shareholder funding
Leverage & pendanaan	maturity profile dan biaya efektif pembiayaan; financial covenant yang membatasi dividen atau corporate action; kebutuhan pembiayaan s.d. akhir tahun dan tahun berikutnya; rencana yang membutuhkan setoran modal
Piutang usaha	aging 0–30, 31–60, 61–90, >90 hari; porsi tertagih setelah tanggal laporan; piutang material yang terlambat atau dispute; penyebab realisasi jauh di atas RKAP
Accrued income	porsi yang sudah menjadi invoice; porsi yang sudah menjadi kas; dasar pengakuan pendapatan; saldo berumur panjang berpotensi koreksi
Utang usaha	konsekuensi ekspansi atau tekanan modal kerja; porsi yang telah jatuh tempo; dampak pada vendor strategis
Perpajakan	timing difference atau lewat jatuh tempo; pemeriksaan, sengketa, denda, bunga; exposure material lain
Hak ekonomi YKPP (P1)	jadwal pembayaran utang dividen; sumber kas pembayaran; porsi YKPP; kebijakan dividen ke depan

12. Drafting guardrails for the LLM

- Input to the writer: template (appendix A), formatted metric strings, tier assignments with rule IDs, selected questions, and management explanations quoted from the RKAP sheet. Nothing else.
- The writer may not compute, round or convert numbers; it may only place supplied strings.
- The writer may not add topics, tiers or recommendations to PT POJ management.
- Forbidden: imperative instructions to PT POJ, definitive judgements of wrongdoing, any figure without "sekitar".
- After drafting, a checker extracts every number from the text and matches it against the metric file. Any orphan number fails the draft and triggers a regeneration with the orphan listed.

- If V-04 failed, growth sentences are replaced by a neutral statement that periods are not comparable.

13. Success metrics

Type	Metric	Target	Measured by
Leading	P0 extraction accuracy on fixtures	≥ 99,5% of fields exact	Automated test vs section 8
Leading	Orphan numbers in drafts	0	Grounding checker log
Leading	Time from upload to draft	≤ 15 minutes	Run logs
Lagging	Tier agreement with analyst	≥ 90% (stretch 95%)	Override log over 3 cycles
Lagging	Analyst edit time	≤ 30 minutes	Self-reported per memo
Lagging	Source issues raised to PT POJ that the agent found first	Tracked, no target in v1	Issue log

14. Risks and mitigations

Risk	Impact	Mitigation
Scanned pages with signatures and stamps degrade extraction	Wrong figures	Vision extraction plus V-01 / V-02 / V-05; low-confidence fields shown to analyst
RKAP arrives late or in varying formats	A third of the memo blocked	Graceful degradation (story 6); fixed RKAP template agreed with PT POJ (Q1)
Thresholds tuned to one sample	Wrong tiering	Shadow mode for 3 cycles; analyst overrides feed recalibration
Narrative overreaches governance limits	Reputational, governance	Guardrails in section 12; human sign-off is mandatory
Confidential FS sent to an external model	Data governance	Decision required before build (Q7)

15. Open questions

#	Question	Owner	Blocking
Q1	Where does RKAP data come from and in what format? Can PT POJ send a fixed template every month?	Finance PT POJ / Analyst	Yes
Q2	Did PT POJ's August package itself show July 2025 comparatives, or was that introduced while preparing the memo?	Analyst	Yes (defines V-04 behaviour)
Q3	Definition of total pinjaman: Note 16 only, or including Kewajiban Pihak Berelasi Rp100 M? (Aug memo: Rp727,97 M)	Analyst	Yes
Q4	Sign-off on initial tier thresholds	Kepala Divisi	No
Q5	Should the Hak Ekonomi YKPP module appear in the memo body or as an annex?	Pengurus YKPP	No
Q6	Are past memos and analyst time records available for calibration and a baseline?	Analyst	No
Q7	May the financial statements be processed by an external LLM API?	IT / Legal	Yes
Q8	Source for the related-party vs third-party split of receivables (Note 5 combines them)	Finance PT POJ	No

16. Phasing

Phase	Weeks	Scope	Exit criterion
0. Fixtures	1–2	Collect 6+ input / output pairs; build reference values like section 8; seed error cases	Fixture set signed off
1. MVP (P0)	3–6	FR-01 to FR-09 for PT POJ	All fixture tests pass
2. Shareholder modules (P1)	7–10	FR-10 to FR-14, review UI	Analyst accepts evidence view
3. Shadow run	3 monthly cycles	Agent drafts in parallel with the manual memo	≥ 90% tier agreement
4. Go-live	After phase 3	Agent draft becomes the starting point	Kepala Divisi approval

Appendix A — Template text with placeholders

Fixed blocks reproduced from the sample memo with variables in braces. These are passed to the writer verbatim.

Paragraf 1 — Purpose

Sehubungan dengan kepemilikan saham YKPP pada PT Pesonna Optima Jasa (PT POJ) dan telah tersedianya Laporan Keuangan PT POJ periode {BULAN_TAHUN}, Divisi Keuangan & Subsidiary telah melakukan analisis terhadap perkembangan kinerja keuangan PT POJ sebagai bagian dari monitoring atas penyertaan langsung YKPP.

Paragraf 2 — Positioning (fixed)

Dalam melakukan analisis tersebut, perlu ditempatkan secara proporsional bahwa YKPP berkedudukan sebagai pemegang saham non-pengendali PT POJ. Oleh karena itu, monitoring yang dilakukan YKPP tidak dimaksudkan untuk memasuki kewenangan pengurusan dan pengendalian operasional PT POJ, melainkan untuk memperoleh informasi yang memadai mengenai perkembangan nilai investasi, kualitas kinerja perusahaan, potensi return kepada pemegang saham, serta risiko yang dapat memengaruhi kepentingan ekonomi YKPP. ...

Paragraf 3 — Headline

Secara umum, kinerja PT POJ sampai dengan {BULAN_TAHUN} {NADA_KINERJA}. Pendapatan mencapai sekitar {M01}, {ARAH} sekitar {M02} dibandingkan periode pembandingan sekitar {M01_C}. Laba tahun berjalan juga {ARAH_NI} dari sekitar {M03_C} menjadi {M03}, atau tumbuh sekitar {M04}.

Paragraf 4 — Pivot (fixed)

Meskipun demikian, pertumbuhan pendapatan dan laba tersebut perlu dibaca secara utuh bersama perkembangan posisi keuangan PT POJ. ...

Paragraf 5 — Tier legend (fixed)

Untuk memudahkan pemantauan, hasil analisis dikelompokkan berdasarkan tingkat perhatian Merah, Kuning, dan Hijau, dengan kondisi Merah ditempatkan sebagai perhatian utama.

Kesimpulan — Merah recap

Kondisi yang dikategorikan Merah menjadi area perhatian utama dalam monitoring selanjutnya, khususnya terkait {DAFTAR_TOPIK_MERAH}. ...

Appendix B — Expected draft item from the July 2026 package

What the agent should produce for Merah item 1 using the July fixture. Every figure maps to section 8.

1. Likuiditas dan Negative Working Capital

Per 31 Juli 2026, aset lancar PT POJ tercatat sekitar Rp389,88 miliar, sedangkan liabilitas lancar mencapai sekitar Rp653,35 miliar. Kondisi tersebut menghasilkan current ratio sekitar 0,60x dan negative working capital sekitar Rp263,47 miliar, melebar dibandingkan sekitar Rp152,84 miliar pada periode pembandingan.

Bagi YKPP, kondisi tersebut perlu dipahami untuk mengetahui apakah struktur modal kerja tersebut merupakan karakteristik normal dari model bisnis PT POJ atau telah menimbulkan ketergantungan terhadap refinancing dan tambahan pembiayaan, terlebih dengan adanya utang dividen sekitar Rp110,27 miliar yang tercatat dalam liabilitas lancar.

Pertanyaan kritis: bagaimana manajemen menilai kecukupan likuiditas PT POJ untuk memenuhi kewajiban selama 12 bulan mendatang; bagaimana maturity profile pinjaman jangka pendek sekitar Rp399,47 miliar; serta kapan dan dari sumber kas apa utang dividen akan dibayarkan kepada pemegang saham.

Expected July tiering: Merah — likuiditas, leverage, accrued income, utang usaha, perpajakan (piutang vs RKAP cannot be assessed; flagged in the issue log). Kuning — gross margin drift, asset intensity, borrowing-cost anomaly. Hijau — revenue and profit growth, net margin improvement, CFO / net profit 1,50x.

PRD — Automated Verification of Subsidiary Management Reports (Laporan Manajemen → Nota Dinas)

📅 5 Oct 2026 · @Blend Warrior

Overview

We will build a pipeline that turns a subsidiary's monthly **Laporan Manajemen** (scanned PDF) into a draft **Nota Dinas Hasil Verifikasi** (DOCX) for YKPP's Divisi Keuangan & Subsidiary, in the same structure, tone and severity scheme as the reference output.

Context. Each month PT Era Permata Sejahtera (PT EPS) sends YKPP (its parent, Yayasan Kesejahteraan Pegadaian Permata) a 23-page management report: balance sheet, P&L, account notes (CALK), budget realisation and per-unit P&L. An analyst reads it and writes a 4–6 page Nota Dinas to the Pengurus that summarises performance, flags anomalies, requests evidence and classifies findings as Major, Medium or BPR.

Problem. The work is manual, slow and inconsistent. The August 2026 reference Nota Dinas is strong on narrative but missed several hard arithmetic and cross-reference breaks that a machine check catches in seconds (see Validation checks and gap analysis).

Product. Ingest any report with this layout → extract to a canonical data model → run deterministic checks and ratio analysis → generate the Nota Dinas draft with every number traceable to a page and line of the input. A human verifier edits and signs.

Reference pair used for this PRD: `input_pengendali_.pdf` (Laporan Manajemen Agustus 2026, 23 pages, Epson Scan 2 image PDF with no text layer, so OCR is mandatory) and `output_pengendali_.docx` (Nota Dinas, 19 sections, ~22k characters).

Goals, non-goals and success metrics

Goals

1. Accept any monthly or quarterly Laporan Manajemen that follows the PT EPS layout and produce a Nota Dinas draft in under 10 minutes.
2. Extract every figure into a structured, typed dataset with page/line provenance.
3. Run a fixed library of integrity, ratio, variance and classification checks so no arithmetic or cross-reference break is missed.
4. Classify each finding as Major Finding, Medium, Medium/Monitoring or BPR with a stated rule, and map it to a follow-up action.
5. Keep the verifier stance of the reference: findings are “indications needing clarification”, never audit opinions.

Non-goals

- Issuing an audit opinion or posting corrections to PT EPS books.
- Tax advice; the system flags tax-rate and reconciliation questions only.
- Free-form reports from other subsidiaries with a different layout (phase 2, via template mapping).

Success metrics

Metric	Target
OCR field accuracy on numeric cells (after validation loop)	≥ 99.5%
Footing / cross-reference breaks detected vs. manual re-check	100%
Reference findings reproduced (12 items in section XVII)	12 of 12
Additional valid findings beyond reference, accepted by verifier	≥ 5 per report
Every number in Nota Dinas traceable to source page + line	100%
Analyst time per report	from ~1–2 days to < 2 hours review

Input taxonomy (Laporan Manajemen)

The input is a fixed-order bundle of 9 block types across 23 pages; four of them are numeric tables that must reconcile with each other.

#	Block (page)	Type	Key fields	Units	Periods
A	Cover (unnumbered)	Metadata	entity, report period, issuer unit, issue date	–	–
B	Cover letter (unnumbered)	Metadata	letter no., addressees, signer, legal basis (AD Pasal 16 ayat 2), cc list	–	–
C	Neraca (p.1)	Statement	line item, Cat (note ref 3–20), current, comparative, Δ%	Rp full	31-Aug-26 vs 31-Aug-25
D	Laporan Laba Rugi (p.2)	Statement	line item, Cat (21–24), current, comparative, Δ%	Rp full	YTD Jan–Aug 26 vs 25
E	Penjelasan Pos Neraca (p.3–8)	Notes / CALK	note no., sub-lines, subtotal, narrative policy text	Rp full	both years
F	Penjelasan Pos Laba Rugi + PPh Badan (p.8–9)	Notes / CALK	revenue split, expense split, tax computation	Rp full	both years
G	Realisasi Anggaran Perusahaan (p.10–14)	Budget table	No, uraian, Real-2025 (3), Anggaran-2026 (4), Real-2026 (5), %(5:4), %(5:3)	Rp 000	YTD + full-year budget
H	Laba Rugi per Unit Bisnis (p.15–17)	Segment table	Outsourcing, Non-Outsourcing, Kantor Pusat; same 7 columns	Rp 000	same as G
I	Sub-unit Non-Outsourcing (p.18–21)	Segment table	Properti/Ekspedisi, Kredit K3P, Rental, Usaha Lain	Rp 000	same as G

Data model (canonical). Every numeric cell becomes one fact row:

Format conventions the parser must handle

- Indonesian number format: . thousands, , decimals; negatives in parentheses; – means zero.
- Mixed units: statements and CALK in full Rupiah, budget and segment tables in Rp 000.
- Spreadsheet residue: #DIV/0! and blank Δ% cells are “not computable”, not errors to report.
- Column header inconsistency: “31-Aug-26” vs “31 Agustus 2025”, “Agst”; normalise to ISO dates.
- Label drift: the same account is “Pendapatan Jasa Expedisi/Properti”, “Logistik dan Expedisi” and “Pendpt Js Property”; needs a synonym map to a chart-of-accounts code.
- Mis-labels to keep verbatim but flag: P&L last line reads “Laba Bersih **Sebelum** Pajak” although it is after tax; unit tables read “Jumlah”.
- Handwritten signature, logos and stamps: ignore.
- Narrative text in CALK (policies, explanations) is extracted as text and linked to its note number; it is evidence for classification checks.

Output taxonomy (Nota Dinas Hasil Verifikasi)

The output is a Bahasa Indonesia memo with a fixed header, a three-paragraph preamble, 19 Roman-numbered sections and a signature block. Sections fall into five functional groups.

Header and preamble (fixed template)

- Header table: Nomor, Tanggal, Kepada (Pengurus YKPP), Dari (Kadiv Keuangan & Subsidiary), Perihal ("Hasil Verifikasi Laporan Keuangan {entity} Periode {month year}"), Lampiran.
- Para 1: receipt of the report and scope. Para 2: verification approach. Para 3: disclaimer — not an audit opinion, the subsidiary remains responsible for data and evidence.

Section map

Group	Sections	Content rule	Driven by input blocks
1. Performance summary	I	Revenue, opex, operating profit, net profit, YoY growth, operating margin trend; closing line on quality of revenue/earnings	